Polytechnic University of Puerto Rico
Biomedical Engineering Department

Spin It 3D

Electrical Subsystem Connection Guide

Clear wiring plan for the MCU, display, sensors, solenoid lock, and motor-speed control.

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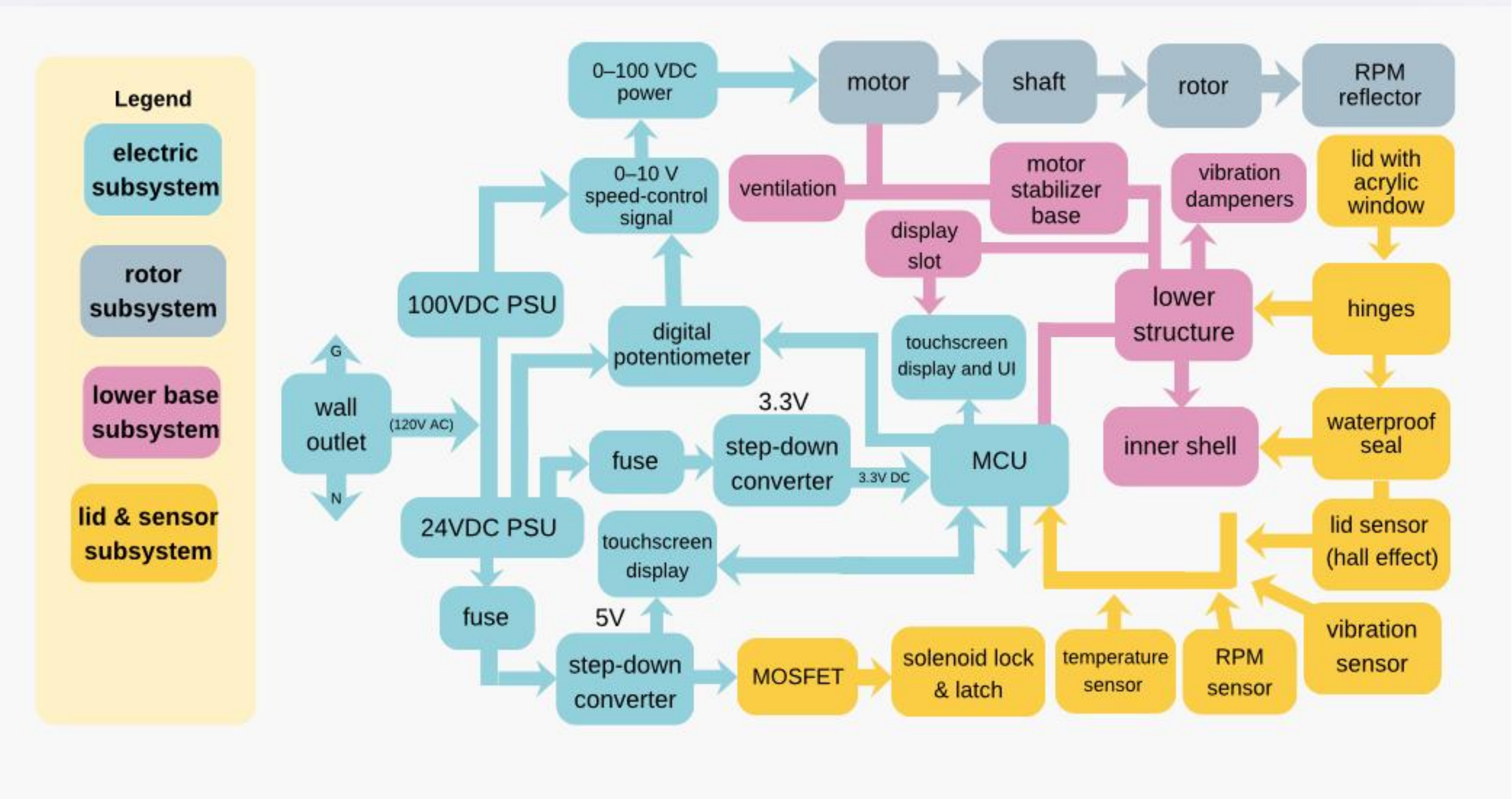
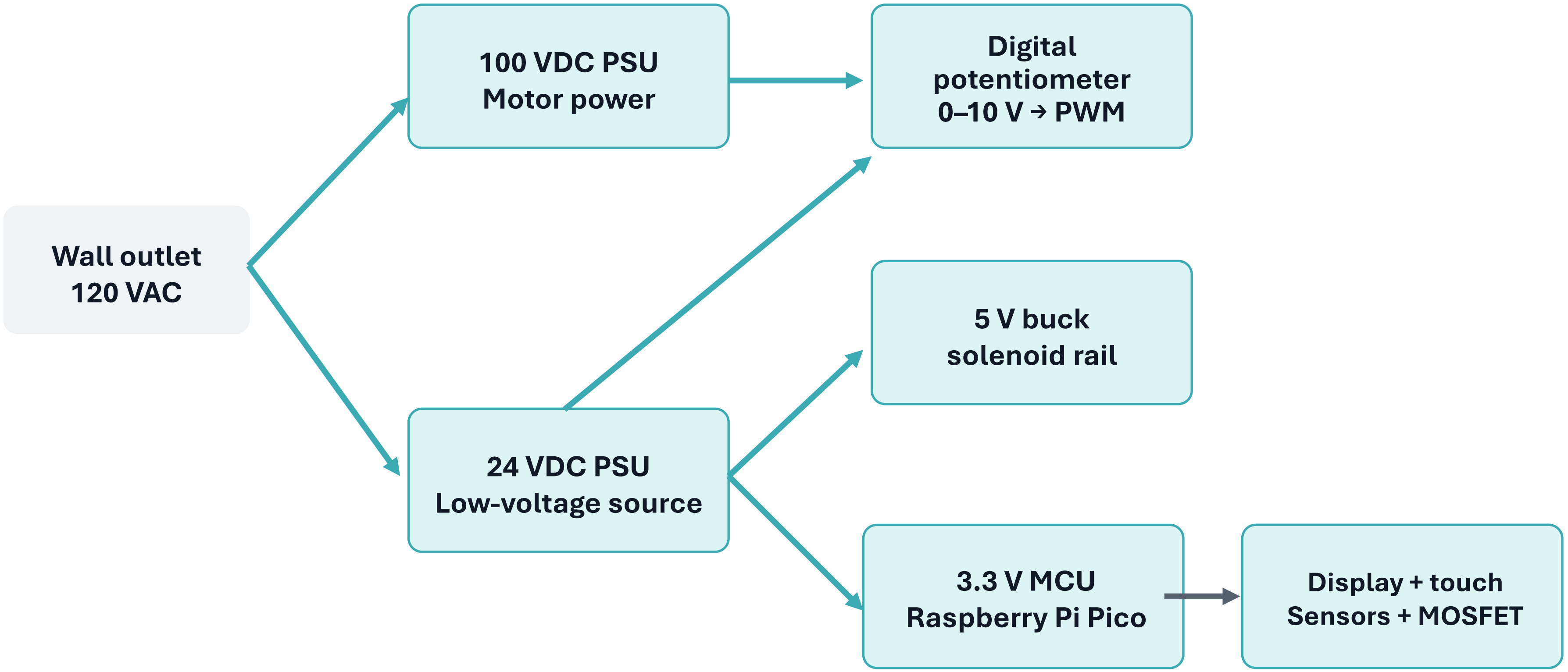


Legend
electric subsystem
rotor subsystem
lower base subsystem
lid & sensor subsystem
0–100 VDC power
motor
shaft
rotor
RPM reflector
0–10 V speed-control signal
ventilation
motor stabilizer base
vibration dampeners
lid with acrylic window
display slot
100VDC PSU
digital potentiometer
touchscreen display and UI
lower structure
hinges
G
wall outlet
(120V AC)
N
3.3V
fuse
step-down converter
3.3V DC
MCU
inner shell
waterproof seal
24VDC PSU
touchscreen display
lid sensor (hall effect)
fuse
5V
step-down converter
MOSFET
solenoid lock & latch
temperature sensor
RPM sensor
vibration sensor

Electrical subsystem overview



Use two power paths: high power for the motor and low power for control electronics.
The MCU commands speed; it should never carry motor current.
All low-voltage modules need a shared ground reference where signals are exchanged.

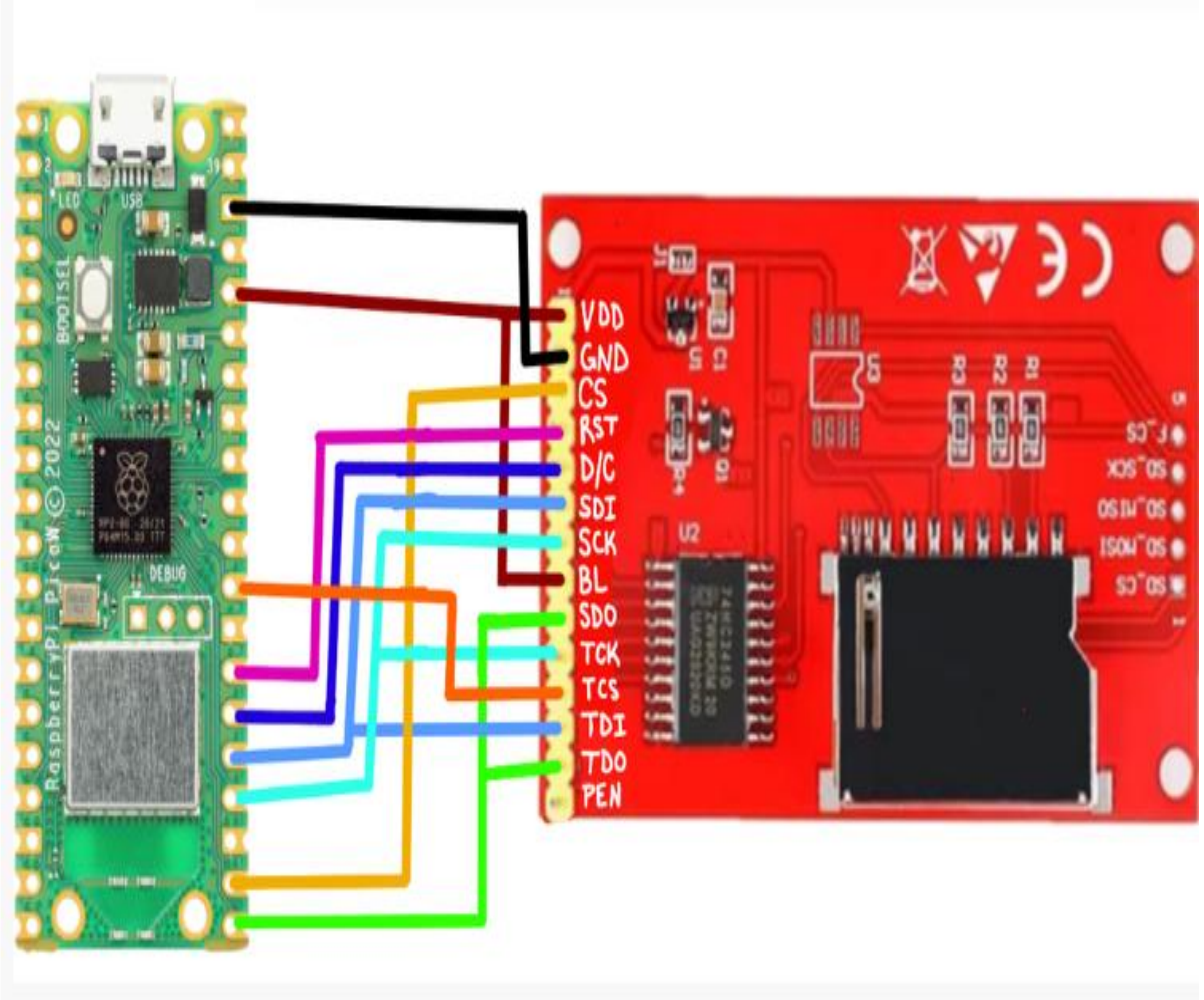
Power distribution and grounding

High-power path	Low-voltage path	Signal reference
120 VAC → 100 VDC PSU → motor and speed controller	24 VDC PSU → buck converters → MCU / solenoid / modules	GND must be common for MCU ↔ sensor/module signals

Put a fuse on each supply branch before it feeds the load.
Do not connect the 100 V motor output to the MCU, sensors, display, or breadboard.
Route motor wires away from sensor and touchscreen signal wires to reduce noise.
Use thicker wire for motor and solenoid current; use signal wire only for GPIO lines.

Key idea: power lines deliver current; GPIO lines only send commands.

Touchscreen display connection

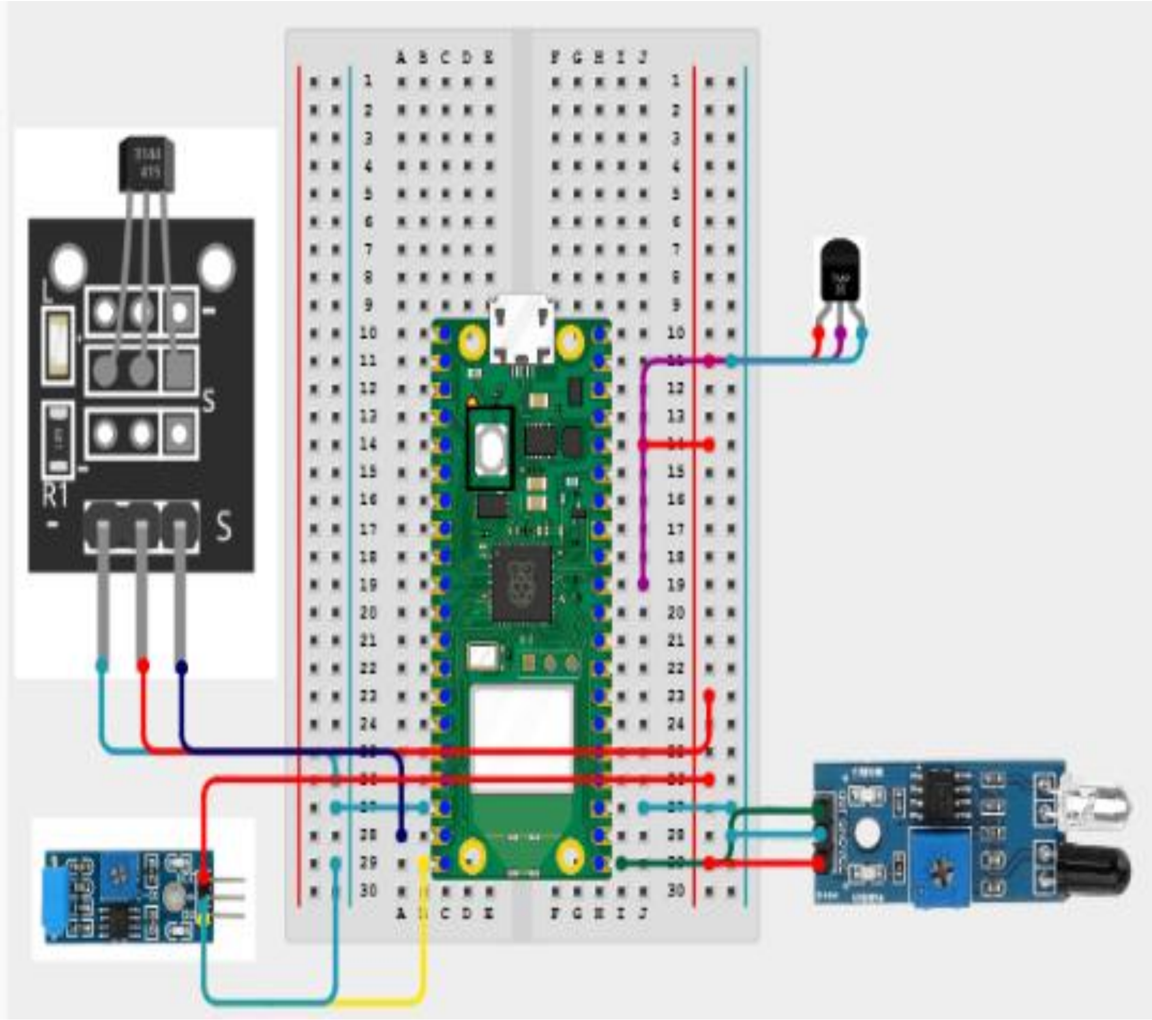


Use 3.3 V logic

VDD → 3V3, GND → common GND, BL → 3V3 or display backlight control.
SPI bus: SCK → GP18, SDI/MOSI → GP19, SDO/MISO → GP16.
TFT control: CS → GP17, D/C → GP21, RST → GP20.
Touch control: TCS → GP22; TDI and TDO share the SPI lines; PEN can stay unused.

In code, keep the display pins grouped in one "pin_config.h" file so changes are easy.

Sensors connection pattern



Each sensor needs 3 lines

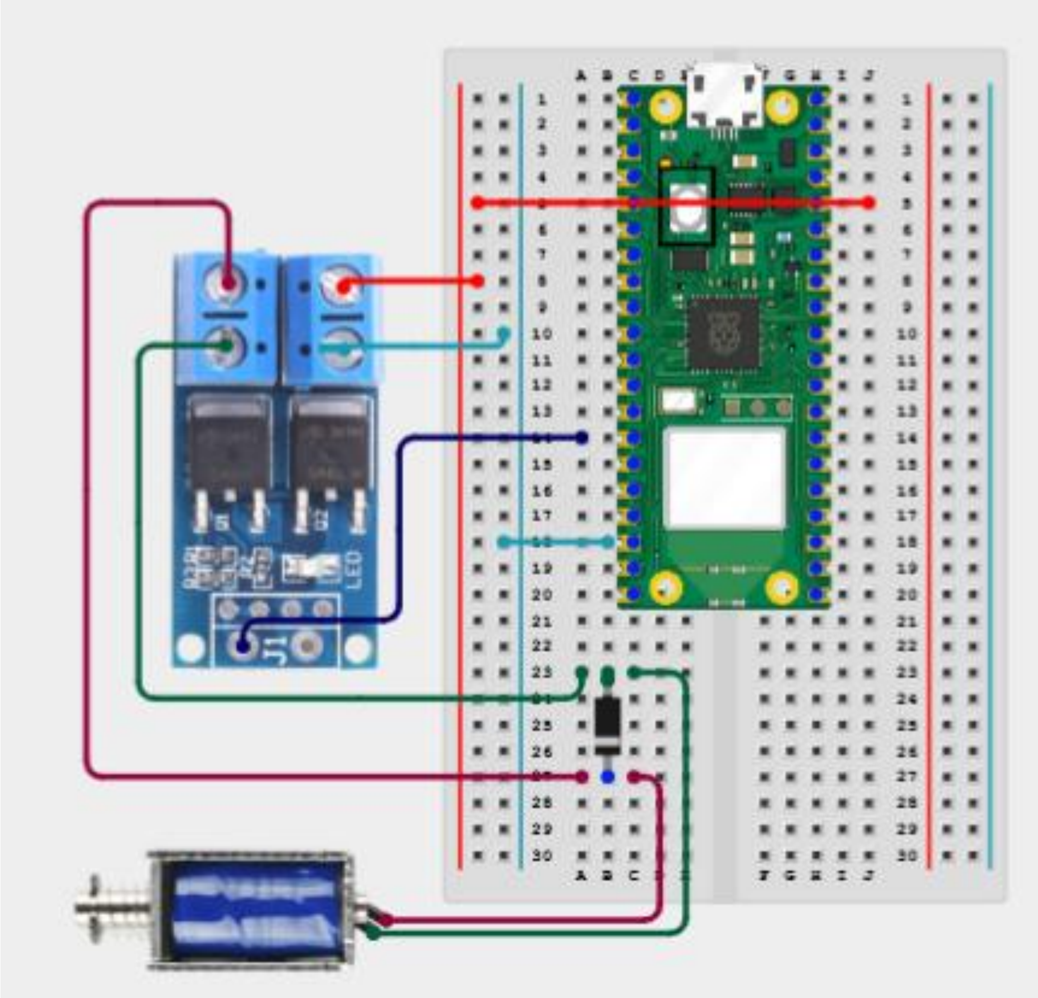
VCC → sensor power

GND → common ground

OUT → GPIO input

Hall lid sensor: OUT changes when the lid magnet is near the sensor.
IR RPM sensor: OUT pulses each time the rotor reflector passes.
Temperature and vibration sensors: OUT goes to analog or digital GPIO depending on the module.

Solenoid lock connection

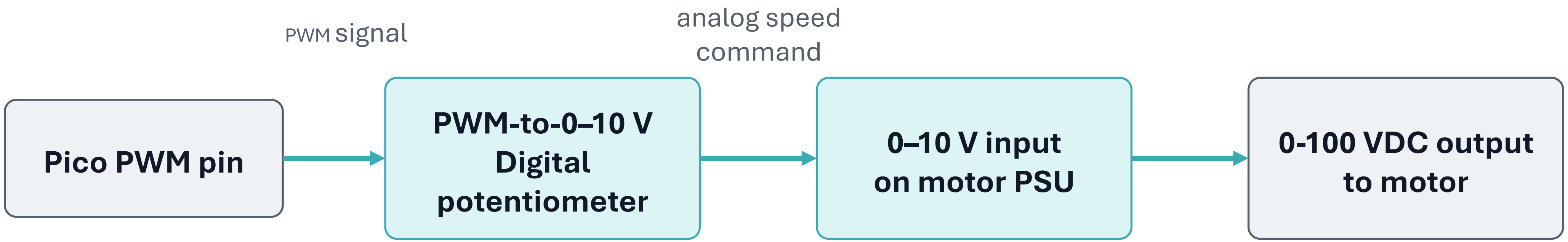


Solenoid wiring

+5 V buck output → solenoid positive lead.
Solenoid negative lead → MOSFET drain.
MOSFET source → 5 V supply ground.
GPIO → 220 Ω resistor → MOSFET gate; add 10 kΩ gate pulldown to GND.
Flyback diode across solenoid: stripe side to +5 V.

The Pico pin only switches the MOSFET gate. The solenoid current comes from the 5 V buck supply, not from the Pico.

Motor speed-control connection



Power the converter from the 24 V rail if the module accepts 12–30 V.
Connect converter OUT+ and GND/OUT− to the PSU's 0–10 V speed input terminals.
Calibrate the module so 0% PWM ≈ 0 V and 100% PWM ≈ 10 V.
Keep the 100 V motor output isolated from the MCU and breadboard.

Bring-up checklist

1. Verify each power rail with a multimeter before connecting loads.
2. Power the Pico + display only; confirm the UI loads correctly.
3. Add sensors and check that each GPIO changes state in serial monitor.
4. Test the solenoid with a short timed pulse before installing it on the lid.
5. Test motor speed control with no rotor first, then with a balanced rotor and enclosure.

Never test a high-speed rotor without containment, lid interlock, and emergency stop available.